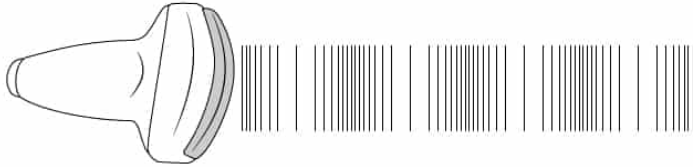


An ultrasound probe emits sound waves that displace molecules in a medium as they travel away from the probe, as shown in the figure below. Which of the following best describes the waves created by the ultrasound probe?



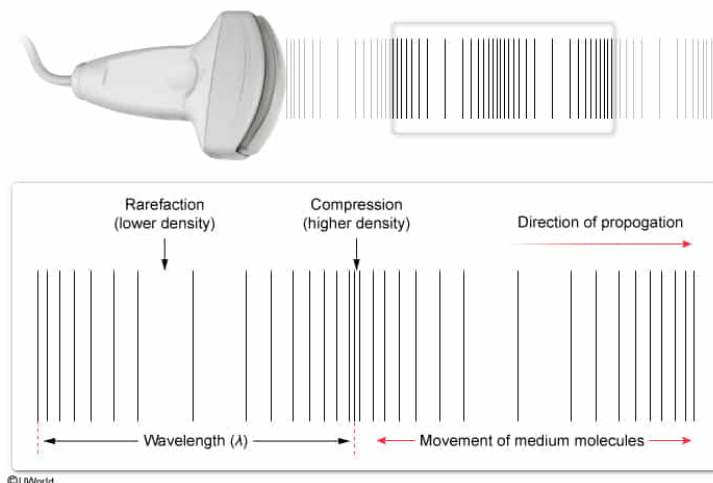
- ✓ ☒ A. Longitudinal waves [76%]
- ☐ B. Low pitch waves [1%]
- ☐ C. Standing waves [8%]
- ☐ D. Transverse waves [12%]

Correct

77% answered correctly

Explanation:

Longitudinal wave produced by an ultrasound probe



A **wave** is a motion created by a disturbance in a medium. Mechanical waves displace molecules and propagate through a medium in two different wave forms: **transverse and longitudinal**.

Transverse waves displace the components of the medium perpendicular to the direction of the wave's propagation. An example of a transverse wave is a water wave in the ocean, where the water

risers and falls vertically relative to the ocean surface, but the wave travels horizontally parallel to the surface.

In contrast, **longitudinal waves** displace the medium in a direction parallel to that of the wave's propagation. For example, **sound waves** create disturbances in the air, causing air molecules to oscillate back and forth parallel to the direction the sound wave travels through the air.

In this question, the probe emits **ultrasound waves** (ie, sound waves of **high frequency**), which displace the molecules in the medium along the direction of the traveling wave. Therefore, the ultrasound waves produced by the probe are longitudinal waves.

(Choice B) Low pitch sound has a lower frequency. However, ultrasound waves are sound waves of high frequency (ie, greater than 20 kHz), which cannot be heard by humans.

(Choice C) Standing waves are created by an interference pattern of reflected and transmitted waves inside a fixed region.

(Choice D) Transverse waves displace the medium in a direction perpendicular to the traveling wave.

Educational objective:

Longitudinal waves such as sound waves displace a medium parallel to the direction of the wave's propagation. Ultrasound waves are sound waves of high frequency.

Time spent:0 sec

Subject:
Physics

QID:403957

System:
Light and Sound